Methods in genome annotation



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Free as in Freedom

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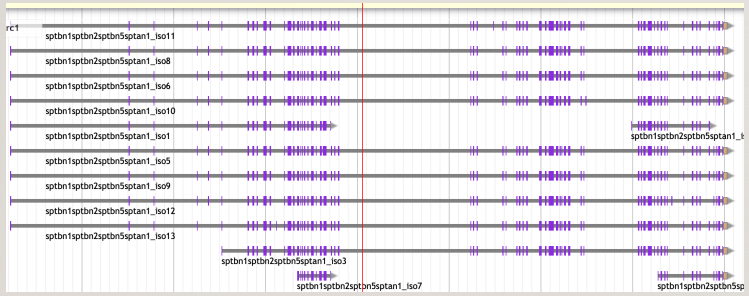


What is annotation ?

Structural annotation:

Find out where the regions of interest (usually genes) are in the sequence data and what they look like.

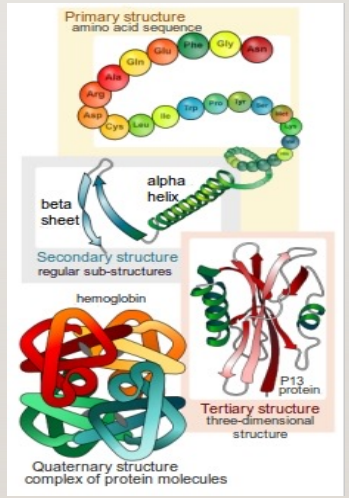
=> **Gene prediction** / **Gene Finding**



vs

functional annotation:

Find out what the regions do.
What do they code for?





Annotation tools

First annotation tools

90's

>100 annotation tools*

Nowadays

Structural annotation + functional annotation + …

GFF

Experimental vs Preditive

Experimental
(ESTs, cDNAs, RNA-seq)

Isolate and clone cognate transcripts (as cDNA) sequence them and compare cDNA with genomic DNA

=> It's the ONLY secure method but:

- Cloning is time consuming.
- Lowly expressed genes are difficult to detect.
- …

Predictive

- Intrinsic (Machine Learning)

Use only the DNA sequence itself to predict genes based on statistical models of gene structure

- Extrinsic

Use **external biological evidence**, like RNA-Seq, ESTs, protein alignments, or known homologs, to guide annotation

- Hybrid

Modern methods often integrate intrinsic and extrinsic signals, blurring the line…

The different approaches

- Similarity-based methods :

  These use similarity to annotated sequences like proteins and transcripts

- *Ab initio* prediction :

  Machine learning based methods (probabilistic: Likelihood, Bayesian | deterministic: Neural networks)

- Hybrid approaches :

  Ab initio tools with the ability to integrate external evidence/hints

- Comparative (homology) based gene finders :

  These align genomic sequences from different species and use the alignments to guide the gene predictions

- Chooser / Combiner :

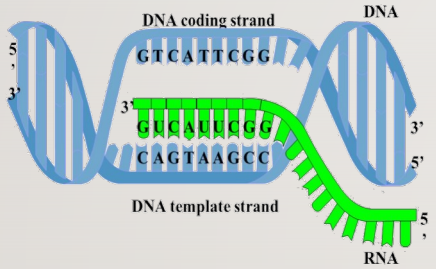
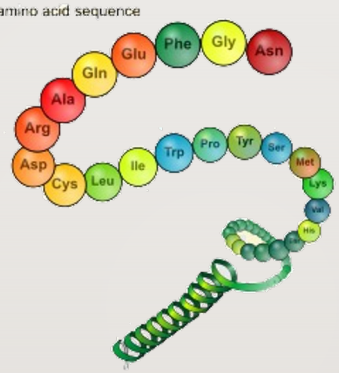
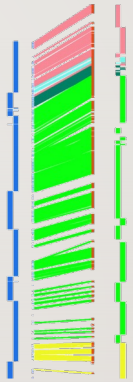
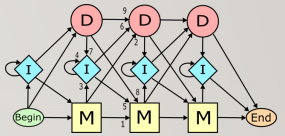
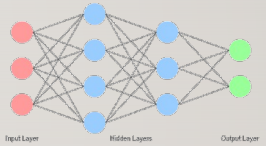
  Selects among existing predictions according to evidence or confidence.

  Constructs consensus gene models by integrating evidence from different sources.

- Pipelines :

  These combine multiple approaches, can annotate other types of features, work on functional annotation, etc.

Types data used vs methods - the main lines

Annotation approach	Extrinsic			Intrinsic
	Transcripts Assembled from RNA-seq or downloaded ESTs 	**Proteins** Known amino acid sequences from other organisms 	**Genomes** Genome from close related species 	**ML models** probabilistic (HMM, GHMM)  deterministic (CNN, RNN i.e deep learning) 
Similarity	X	X		
Comparative			X	
Pure ab initio				X
Hybrid	X	X	X	X
Chooser	X	X		X
Pipeline	X	X		X

Extrinsic approaches

Similarity-based methods

These use similarity to annotated sequences like

Proteins

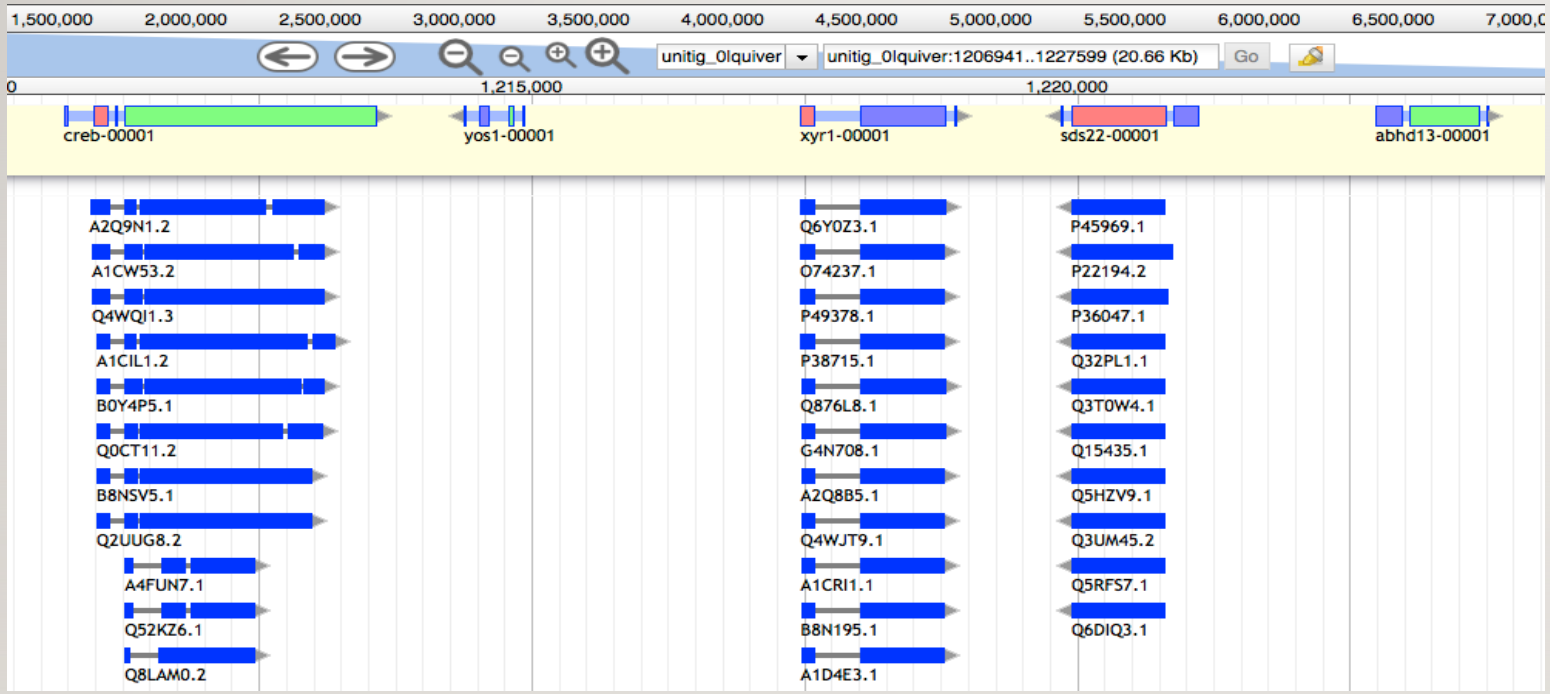
Transcripts

ESTs

Similarity-based method: Protein data

- Evidence sequences (protein, est, etc.) are aligned to the genome

Conserved in sequence => conserved annotation with little noise



Limits :

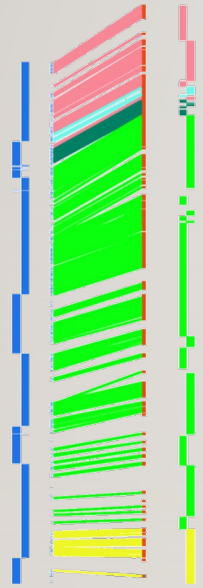
- Related to pre-existing data
- Data from model organisms often used => bias?
- Data can be incomplete
- Data can be wrong

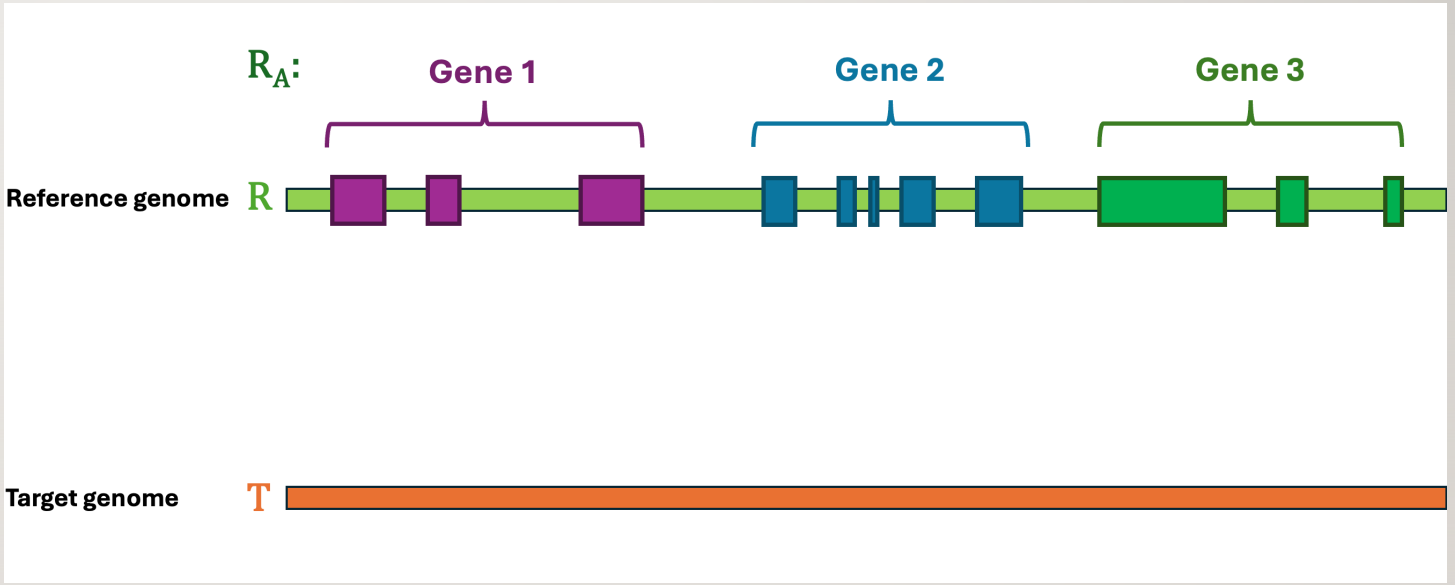
Extrinsic approaches

Comparative methods

Comparative-based method (projection tools)

Tools that use **whole-genome alignments** between closely related species.
Transfer gene models based on conserved synteny, gene structure, and alignment blocks.





LASTZ
Mummer
Satsuma
SynMap
} chained alignments

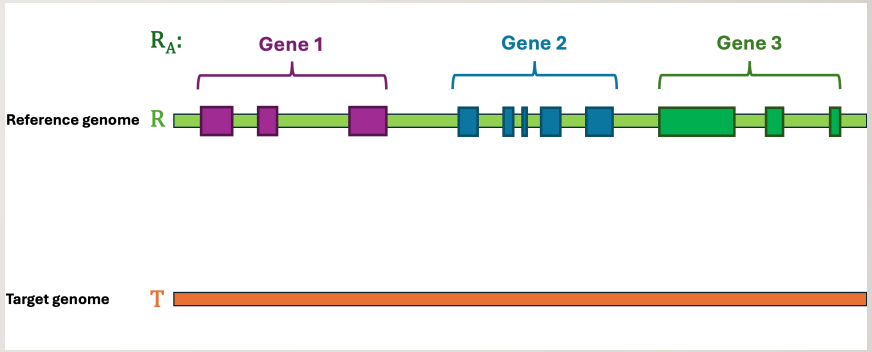
Pecan
HAL
} graph alignments

Popular tools

- **Kraken (not the kmer tool one)**
- **TransMap (UCSC)**

Comparative-based method (projection tools)

Tools that use **local alignments** between species.
Transfer gene models based on gene alignment.



Popular tools

LiftOff

transfer annotation between closely related species

minimap2

LiftOn

transfer annotation between close and distant related species

Miniprot

Intrinsic approaches

ab initio

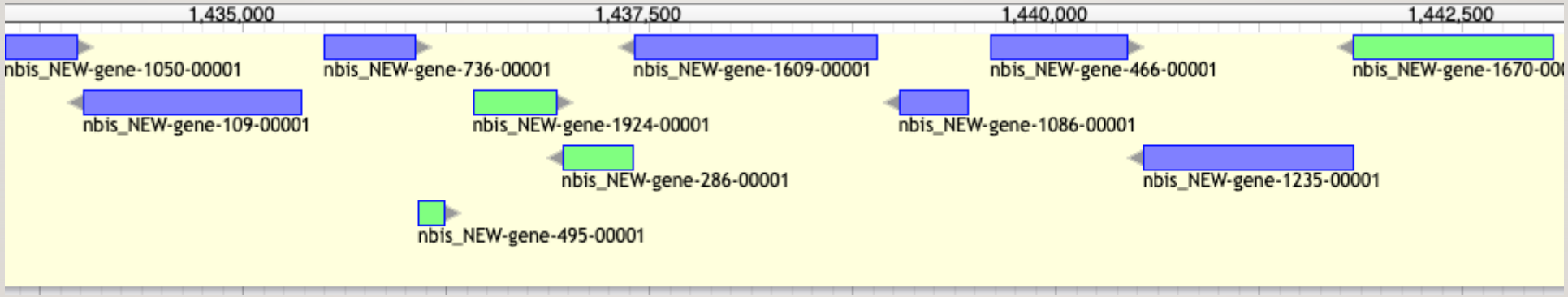
Using Machine Learning approaches (probabilistic and/or deterministic models) to predict features

- **Bacterial gene prediction**

  - Short intergenic regions

  - Uninterrupted ORFs (No intron)

  - Very conserved signals



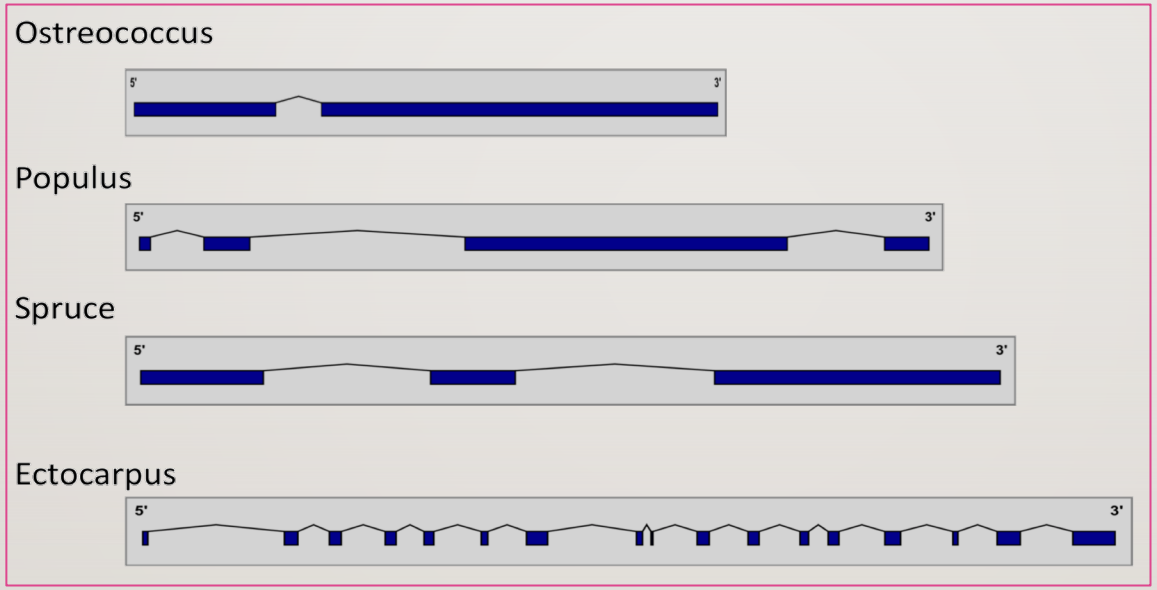
⇒ easy problem: Accuracy > 90%

- **Eukaryotic gene prediction**

  - Presence of intron => structures differ

  - Isoforms

  - New features (e.g. lncRNAs)



=> Hard problem: Accuracy < 70%

=> Requires training

Probabilistic methods

- **gene content**

Statistical properties of protein-coding sequence learn from existing data.

  - codon usage
  - hexamer usage
  - GC content
  - compositional bias between codon positions
  - nucleotide periodicity
  - exon/intron size
  - …

- **signal detection**

These are like **landmarks** in the genome, they tell the model *where* to look.

  - Promoter
  - ORF
  - Start codon
  - Splice site (Donor and acceptor)
  - Stop codon
  - Poly(A) tail
  - CpG islands
  - …

=> *Ab initio* tools will combine this information through different Probabilistic models (e.g. HMM, GHMM, WAM) to calculate the most probable combination that represents a valid gene

Popular tools (Prokaryote)

Complete bacterial or archaeal genomes

- **Prodigal** a widely used baseline for protein-coding gene prediction
- **Pyrodigal a** reimplementation of Prodigal in Python
- **GeneMarkS-2**
- **GLIMMER 3** (requires a training step for a new organism)

Metagenomes assemblies

- **FragGeneScan** (particularly relevant when sequences contain sequencing errors or frameshifts)
- **MetaGeneMark**
- **MetaGene**
- **MetaGeneAnnotator**

Deterministic methods

Neural networks (CNN, bLSTM) predicts **for each base**, the probability of being:

- in an exon
- in an intron
- at a splice site
- in UTRs, etc.

Captures complex, long-range sequence dependencies.
=> struggles to precisely locate exon boundaries
=> base-level predictions can be

- **Noisy** (e.g., false positives),
- **Biologically inconsistent** (e.g., a stop codon in the middle of a predicted CDS),
- **Missing structure** (e.g., incorrect start/stop, fragmented exons).

+

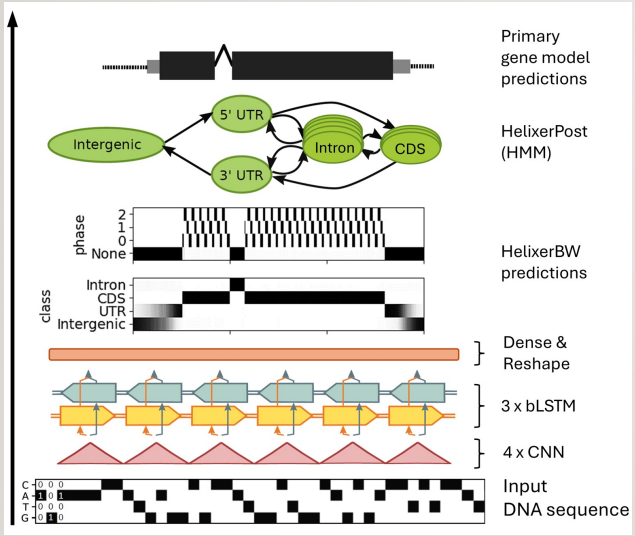
HMM postprocessor

Ensures coherent gene structures by filtering noisy outputs.

Popular tools (Eukaryote)

Helixer

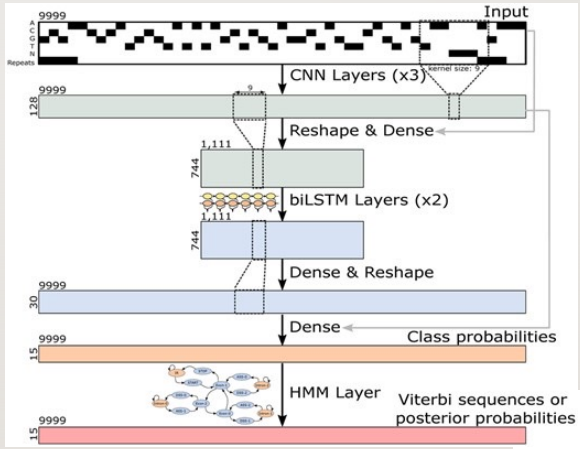
Helixer–de novo Prediction of Primary Eukaryotic Gene Models Combining Deep Learning and a Hidden Markov Model

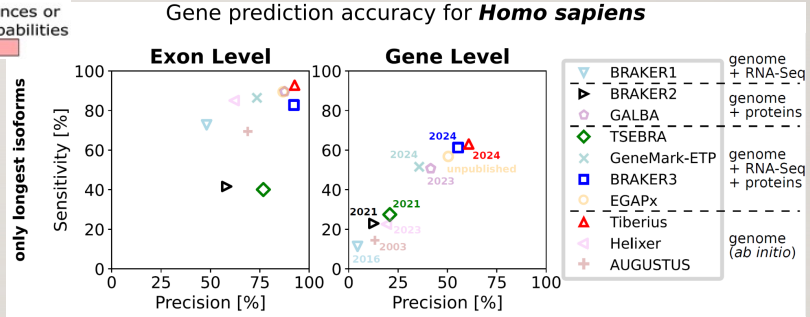


https://doi.org/10.1101/2023.02.06.527280
https://github.com/weberlab-hhu/Helixer

Tiberius

End-to-End Deep Learning with an HMM for Gene Prediction

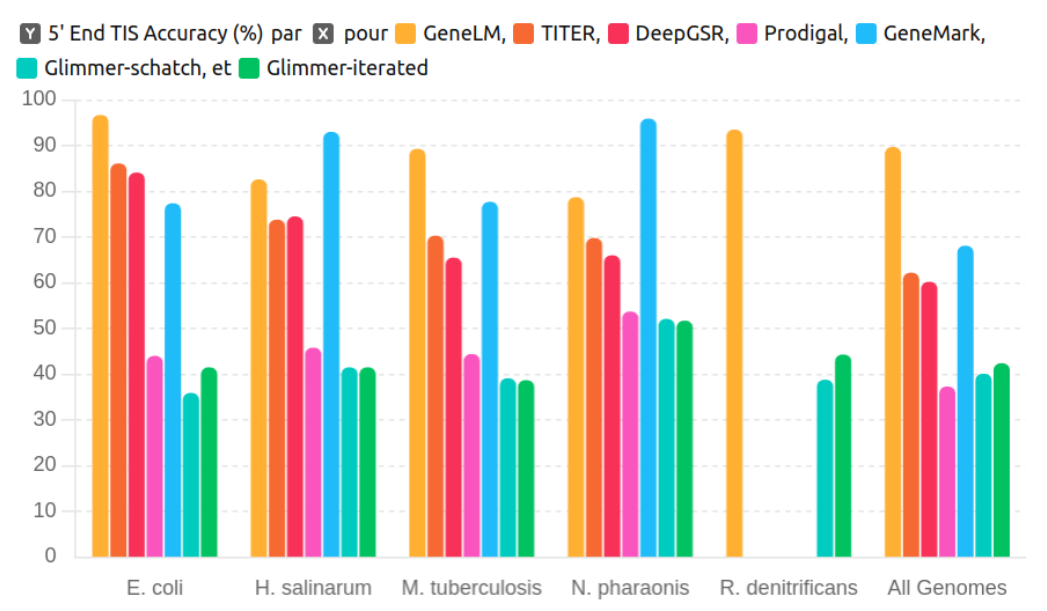




https://academic.oup.com/bioinformatics/article/40/12/btae685/7903281
https://github.com/Gaius-Augustus/Tiberius

Popular tools (prokaryote)

GeneLM



=> needs training

Strengths :

- Fast and easy
- Annotate unknown genes
- Sensitivity ok
- Need no external evidence

Limits :

- Split single gene into multiple predictions
- Fused with neighboring genes
- May be less accurate than homology-based methods (Exon boundaries)
- Some tools need training

Hybrid approaches

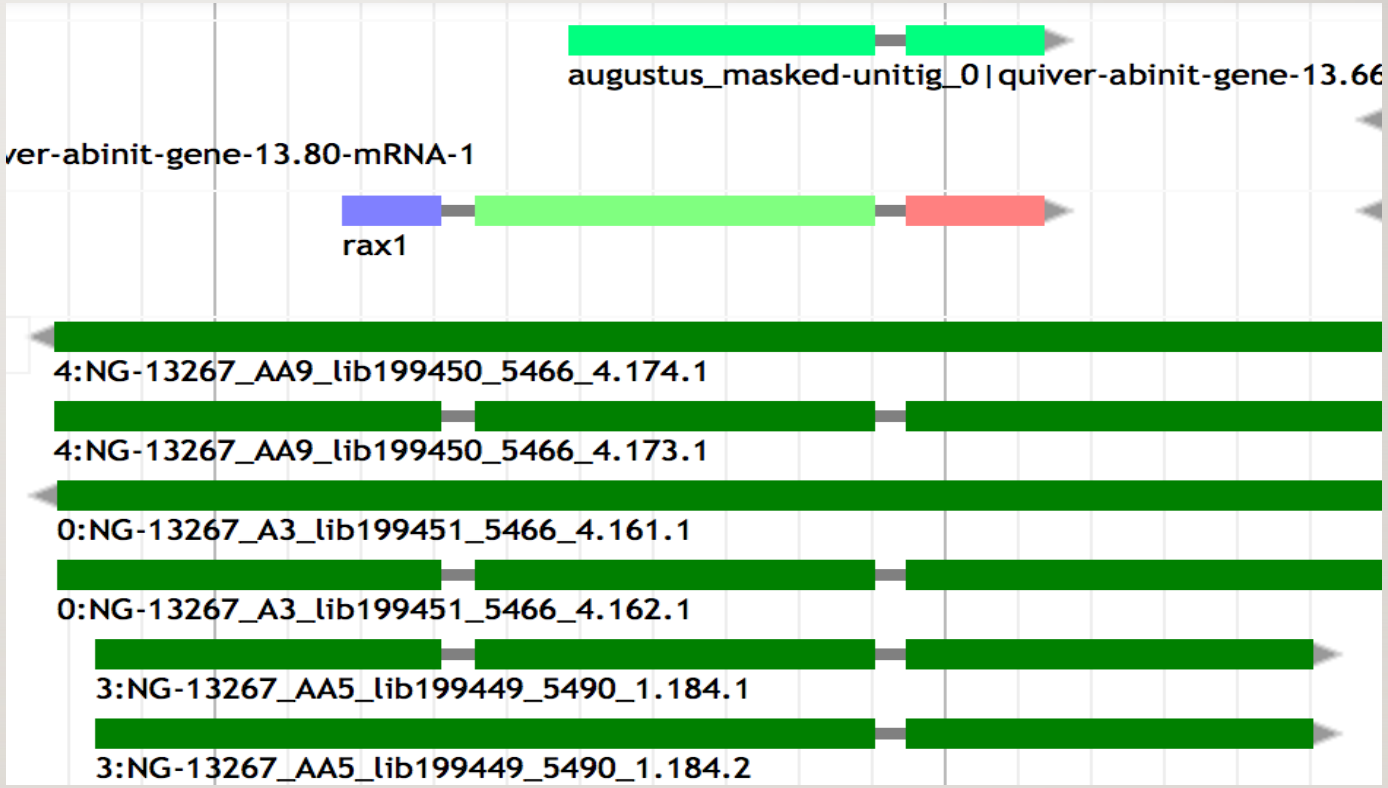
Evidence-drivable gene predictors incorporate hints/extrinsic information in the form of alignments (transcript, protein, whole genome) to increase the accuracy of the gene prediction.

- *Can predict locus without extrinsic information*
- *Improve prediction when extrinsic information available*

Hybrid: Similarity based + *ab initio*

Evidence-drivable gene predictors / Ab initio evidence-based approaches incorporate hints (e.g. protein) alignments to increase the accuracy of the gene prediction.

- **Can improve :**
  - Split single gene into multiple predictions
  - Fused with neighboring genes
  - Exon boundaries
- **Tools :**
  - PGAP (NCBI) - constrains or influences gene prediction
  - Bakta - gene boundaries refinement



Chooser / Combiner

- *Selection of gene models*
- *Combining heterogeneous data into gene models*

Use battery of gene finders and evidence (transcript, protein) alignments and:

Chooser: Selects among existing predictions according to evidence or confidence.

Combiner: Constructs consensus gene models by integrating evidence from different sources.

Tool	Consensus based chooser	Evidence based chooser	weight of different sources	Comment
GeMoMa		X		You can weight on presence in multiple genome
JIGSAW	X			
EVM Evidencemodeler	X	X	X	User can set the expected evidence error rate manually or/and learn from a training set
Evigan	X		X	Unsupervised learning method
=> Ipred		X		Does not require any a priori knowledge Can also combine only evidences to create a gene model

Pipelines

(The ultimate step)

- *Align evidence (proteins)*
- *Annotation of other features (Repeat, tRNA, etc)*
- *Add functions*
- *And more…*

Tool	Protein	*Comparative*	*Ab inito*	Function	Other annotated features	Comment
PGAP	X	X	X	X	tRNA, rRNA, ncRNA, pseudogenes, CRISPR, mobile elements	Integrates protein alignments, protein-family HMMs and GeneMarkS-2+ for structural and functional annotation
Prokka	X		X	X	tRNA, tmRNA, rRNA, ncRNA, CRISPR; signal peptides when supported by the relevant dependencies	Uses Pyrodigal for CDS prediction.
Bakta	X		X	X	tRNA, tmRNA, rRNA, ncRNA, regulatory regions, CRISPR, pseudogenes, replication origins, sORFs	Uses Prodigal for CDS prediction. Huge database used.
DFAST	X		X	X	RNA genes and other genomic features, depending on the modules used	
RAST / RASTtk	X		X	X	RNA features, metabolic functions and other features, depending on the workflow	
BV-BRC	X		X	X	tRNA, rRNA, repeat regions, prophages, antimicrobial resistance genes, virulence factors, protein families	CDS prediction uses Prodigal followed by Glimmer


The End
Thank you for your attention